

Reconfigurable Optical Routing Using Microring Resonators with Interferometric Arms

Martin Skönderas^(1*), Hilmi Othman⁽¹⁾, Kyle R. H. Bottrill⁽¹⁾, David J. Thomson⁽¹⁾, Periklis Petropoulos⁽¹⁾

⁽¹⁾ Optoelectronics Research Centre, University of Southampton, Southampton, SO17 1BJ, UK, m.skenderas@soton.ac.uk.

Abstract We demonstrate energy-efficient, resonance-stable, reconfigurable optical power routing in a microring resonator with interferometrically controlled coupling. The device achieves up to 20 dB power extinction ratio and error-free 80 Gb/s transmission, establishing a compact building block for scalable reconfigurable silicon photonic interconnects. ©2026 The Author(s)

Introduction

Reconfigurable optical routing and power distribution are key functionalities in emerging optical interconnects, particularly in data centre networks where flexibility, scalability, and energy efficiency are critical. The rapid growth of data traffic has driven the deployment of optical switching within data centre fabrics, requiring compact and low-power photonic devices capable of dynamically redistributing optical power while preserving signal integrity at high data rates [1, 2]. These requirements are especially relevant to O-band short-reach interconnects, with emerging opportunities in scalable optical access networks like PONs.

Silicon photonics has enabled versatile switching via cascaded Mach-Zehnder interferometer (MZI) meshes, such as universal linear optical circuits, offering full amplitude/phase programmability. However, these require $O(N^2)$ tuning elements for N ports [3], yielding large footprints and high power (e.g., >1 pJ/bit for 64×64 switches), hindering scalability [4].

Microring resonators (MRRs) provide a compact, low-power alternative benefiting from resonant enhancement. However, conventional add-drop MRRs are intrinsically limited by a coupling coefficient that is fixed by the waveguide geometry during fabrication. As a result, thermal tuning primarily modifies the optical phase, shifting the resonance wavelength. The resulting variation in power distribution between ports arises from detuning relative to resonance, rather than direct control of the coupling condition. This restricts their ability to achieve continuous and reconfigurable power routing, as the power transfer is fundamentally governed by the fixed coupling regime [5, 6].

In this work, we investigate a compact photonic routing element based on an MRR with interferometric arms in the coupling region. Similar configurations have previously been explored for applications such as tunable filtering, Q-factor control, and microwave photonics [7–10]. However, their potential as reconfigurable $1 \times N$ power routing elements has not been explored. In our implementation,

thermo-optic tuning of the interferometric arms enables control of the effective coupling coefficient, allowing redistribution of optical power between output ports without shifting the resonance wavelength, thereby combining the compactness and efficiency of MRRs with MZI-like tunability.

We experimentally demonstrate continuous power routing in a 1×2 , single-ring device, achieving extinction ratios up to 20 dB with stable tuning. Experiments with data signals validate communication readiness, with error-free 80 Gbit/s operation and negligible penalty. These results highlight the potential of the proposed architecture as a compact and energy-efficient building block for reconfigurable optical switching in data centre interconnects and emerging co-packaged optics (CPO) architectures [11, 12].

Operation principle and device configuration

The device consists of a reconfigurable add-drop silicon MRR incorporating interferometric arms to enable tunable coupling, as shown in Fig. 1(a). The structure comprises four coupling regions between the bus waveguides and the microring, forming two interferometric couplers. Each interferometric section includes two directional coupling regions ($3.5 \mu\text{m}$ length, 235 nm gap), a ring arm with radius $R = 9 \mu\text{m}$, and a bus arm of length $L_a = 18.5 \mu\text{m}$. The corresponding ring arm length is $L = \pi R/2$, and the path length difference is defined as $\Delta L = L_a - L$.

The interferometric arms introduce a controllable phase difference $\Delta\Phi$, enabling dynamic tuning of the effective coupling coefficient between the bus waveguide and the ring. Phase tuning is achieved via integrated microheaters placed on the straight sections of the interferometric arms, exploiting the thermo-optic effect in silicon. Owing to the relatively large thermo-optic coefficient of silicon ($1.8 \times 10^{-4} \text{ K}^{-1}$) [13], efficient tuning of the coupling condition can be achieved with moderate electrical power. Titanium nitride (TiN) heaters are employed to locally control the refractive index and hence the phase [14].

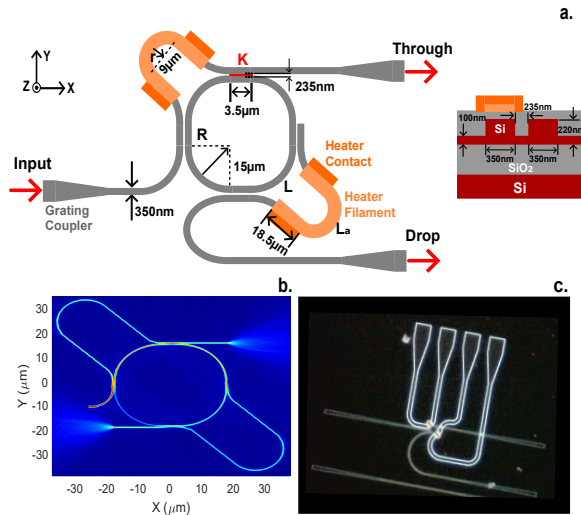


Fig. 1: Device configuration and operating principle of the microring resonator with interferometric arms. (a) Schematic of the device and cross-sectional view of the silicon photonic platform. (b) Simulated optical field distribution (FDTD). (c) Optical microscope image of the fabricated device.

In contrast to conventional add-drop MRRs with fixed coupling coefficients, the proposed architecture enables continuous tuning between under-coupled, critically coupled, and over-coupled regimes. This allows controlled redistribution of optical power between the through and drop ports. Importantly, the tuning mechanism primarily modifies the coupling condition, while inducing only negligible resonance wavelength shift, indicating partial decoupling between coupling strength and phase response.

The waveguides are designed for a silicon rib geometry with a cross-section of $350\text{ nm} \times 220\text{ nm}$ and a slab thickness of 100 nm , supporting only the fundamental transverse-electric (TE) mode. They are designed to operate in the O-band around $1310\text{--}1320\text{ nm}$, matching the low-dispersion window of G.652-based optical interconnects. The optical field distribution, obtained via finite-difference time-domain (FDTD) simulations (Fig. 1(b)), confirms efficient coupling and mode confinement within the interferometric structure and the microring.

The devices were fabricated using the Cornerstone MPW SOI platform (220-nm silicon device layer) with a 120-nm partial etch and a $1\text{-}\mu\text{m}$ SiO_2 top cladding. Integrated metal layers were used to realize the heater filaments and electrical contacts. A microscope image of the fabricated device is shown in Fig. 1(c).

Experimental Results

The fabricated devices were experimentally characterized using an O-band tunable laser and grating-coupler fibre-to-chip interfaces. The output power at the through and drop ports was measured as a function of the current applied to the heater integrated on the upper interferometric arm.

Fig. 2(a) shows the measured optical power at the through and drop ports of a single microring resonator with interferometric arms as a function of heater current. As the applied current increases ($0\text{--}15\text{ mA}$), the device exhibits a clear and monotonic redistribution of optical power between the two ports. At low heater currents ($0\text{--}5\text{ mA}$), the device operates in a drop-dominant regime, where most of the optical power is coupled into the ring and extracted at the drop port with a power extinction ratio of $\sim 20\text{ dB}$. As the heater current increases, the coupling condition is progressively modified, leading to a transfer of power toward the through port.

In the intermediate region ($\approx 11.8\text{ mA}$), both ports carry comparable power, demonstrating fine and continuous control of the splitting ratio. At higher currents ($\approx 14.7\text{ mA}$), the device reaches a through-dominant regime, where the majority of the optical power remains in the bus waveguide. The normalized power traces in Fig. 2(b) further highlight this smooth and controlled transition, confirming conservation of the total power. This behaviour validates the ability of the interferometric arms to effectively tune the coupling condition of the resonator, enabling reconfigurable optical routing within a compact footprint.

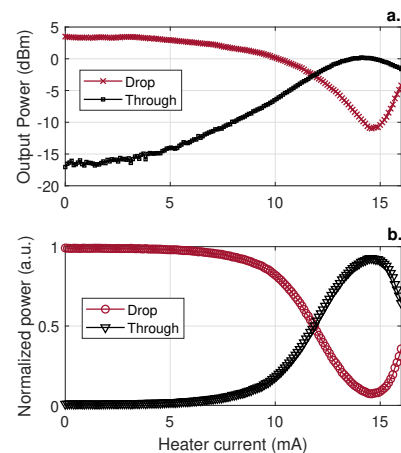


Fig. 2: Measured power routing in a microring resonator with interferometric arms. (a) Output power at the through and drop ports as a function of heater current. (b) Corresponding normalized power.

Fig. 3(a) shows measured transmission spectra of the drop port at different heater currents. Importantly, the resonance wavelength remains nearly unchanged across the full tuning range, with only negligible shifts on the order of a few picometers. In contrast, the spectral line shape undergoes a pronounced evolution. This behaviour confirms that the applied thermal tuning predominantly modifies the effective coupling condition rather than inducing a significant phase shift of the cavity resonance.

This observation is quantitatively supported by Fig. 3(b), where the 3-dB bandwidth is plotted as

a function of heater current. A clear reduction in bandwidth is observed indicating a transition from an over-coupled to a critically coupled regime. Fig. 3(c) shows the evolution of the spectral extinction ratio (ER), defined as the contrast between the resonance peak and the off-resonance transmission (shoulder) level. The spectral ER increases from ~ 20 dB at low heater currents to values exceeding 30 dB in the optimized regime, before dropping sharply at high currents as the device moves away from critical coupling.

Taken together, these results demonstrate that the interferometric arms provide an effective mechanism for reconfigurable control of the resonator coupling condition, enabling simultaneous tuning of power distribution and extinction ratio. A low excess loss of ~ 0.5 dB was evaluated for this design by comparison with reference waveguides.

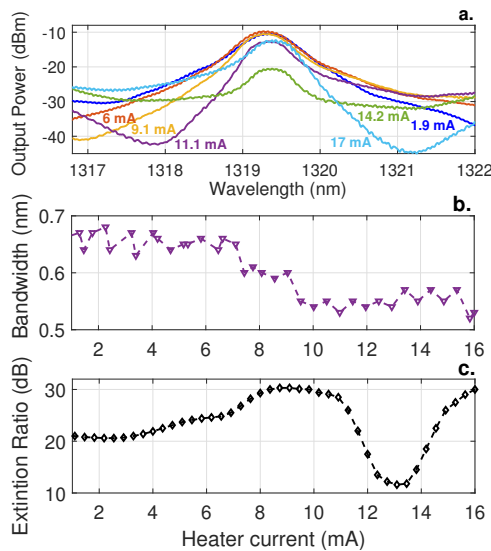


Fig. 3: Spectral response of the microring resonator with interferometric arms. (a) Measured transmission spectra for different heater currents. (b) 3-dB bandwidth as a function of heater current. (c) Extinction ratio versus heater current.

Data Transmission Experiments

To evaluate the suitability of the proposed device for optical communication systems, data transmission experiments were performed using intensity-modulated direct-detection (IM-DD) signals in the O-band. The experimental setup is shown in Fig. 4. A continuous-wave laser source was modulated using an external Mach-Zehnder modulator (MZM) driven by an arbitrary waveform generator to generate 80 Gbit/s PAM-4 signals (40 GBaud - limited by the bandwidth of the MZM available). The modulated signal was polarization-controlled and amplified prior to coupling into the photonic integrated circuit (PIC). At the receiver side, the output signal from the selected port was filtered, attenuated using a variable optical attenuator (VOA) to control the received optical power (ROP), and detected using a photodiode connected to a real-

time oscilloscope for offline processing and bit-error-rate (BER) estimation.

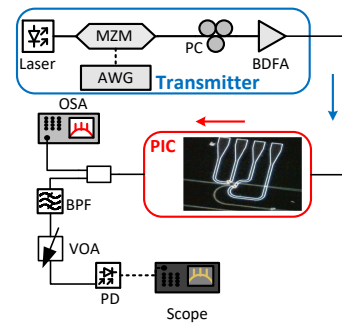


Fig. 4: Experimental setup for data transmission measurements.

Fig. 5(a) shows the measured BER as a function of received optical power (ROP) for back-to-back (B2B), through-port, and drop-port transmission, measured at the heater current corresponding to maximum output power. Both device paths closely follow the B2B reference, with no significant power penalty at the HD-FEC threshold, confirming excellent signal integrity. The close overlap between the ports indicates that the reconfiguration mechanism does not introduce significant port-dependent penalties.

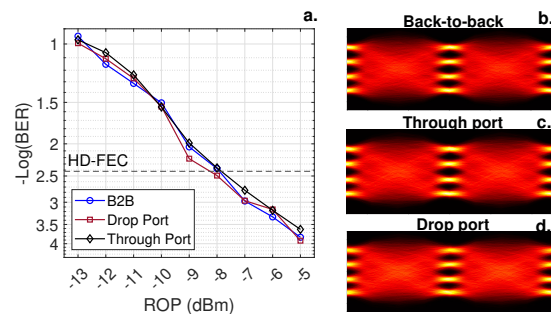


Fig. 5: Data transmission performance of the device. (a) BER as a function of received optical power (ROP) for back-to-back (B2B), through-port, and drop-port transmission. (b–d) Corresponding eye diagrams at ROP = -5 dBm for (b) B2B, (c) drop port, and (d) through port at 80 Gbit/s PAM-4.

Conclusion

We demonstrated a compact silicon photonic routing element based on a microring resonator with integrated interferometric arms, enabling controlled tuning of the coupling condition without significant resonance shift. This allows continuous and deterministic redistribution of optical power across multiple output ports.

Experimental results show power extinction ratios up to 20 dB and insertion loss of ~ 0.5 dB, with consistent power conservation across operating states. Data transmission experiments validate that the device preserves signal integrity with negligible penalty, confirming its suitability for practical optical links and making it a promising building block for dense and programmable photonic switching.

Acknowledgements

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